# Lecture 03: Relationships between 2 variables

# Recap of chapters 1 and 2

Mostly looking at a single variable:

- Graphs to explore the distribution of single variables (histograms, bar charts)
- Summary numbers to describe our distributions:
  - Measures of central tendency (mean, median)
  - Measures of spread (standard deviation, IQR)

# Learning objectives for today

- Time plots to examine what happens to a variable over time
  - using `geom_point()`
  - using `geom_line()`
- Explore the relationship between two quantitative variables
  - Directionality
  - Association vs causation
- Make scatter plots to look at relationships visually
  - using `geom_point()`
- Use the correlation coefficient to quantify the strength of linear relationships
  - calculate correlations using `cor()`

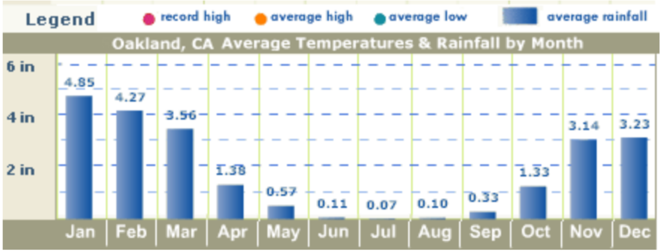
# Time plots

# Visualize quantitative variables over time using time plots

- Time plots are a specific subset of plots where the x variable is time.
- Unlike the previous plots, the time plot shows a relationship between two variables:
  i) a quantitative variable
  ii) time
- Often times, these plots can be used to look for cycles (e.g., seasonal patterns that recur each year) or trends (e.g., overall increases or decreases seen over time).

# Time plot: Oakland rainfall

- from *See California.com*, January 2021:



# Life expectancy for White men in California

Make a scatter plot of the life expectancy for White men in California over time.

Since the dataset contains 39 states across two genders and two races, first use a function to subset the data to contain only White men in California.

Which function from last lecture do we need?

- `mutate()`, `select()`, `filter()`, `rename()`, or `arrange()`?

# filter() to select a subset of rows

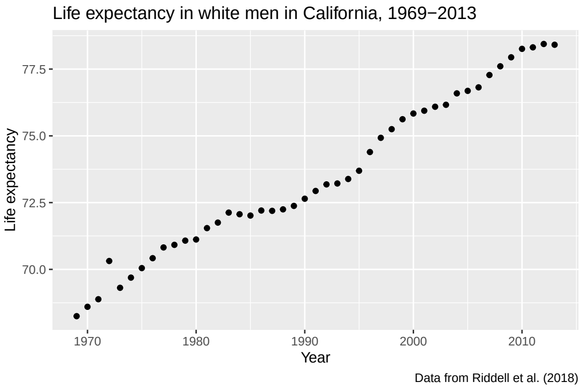
```
wm_cali <- le_data %>% filter(state == "California",
                              sex == "Male",
                              race == "white")

# Equivalent:
wm_cali <- le_data %>%
  filter(state == "California" &
         sex == "Male" &
         race == "white")
```

# Code for `geom_point()`

```
ggplot(data = wm_cali, aes(x = year, y = LE)) +
geom_point() +
labs(title = "Life expectancy in white men in California, 1969-2013",
    y = "Life expectancy",
    x = "Year",
    caption = "Data from Riddell et al. (2018)")
```

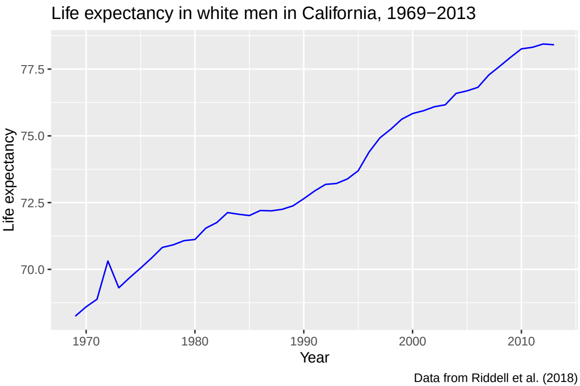
# Point plot: life expectancy over time



# Code for geom_line()

```
ggplot(data = wm_cali, aes(x = year, y = LE)) +
geom_line(col = "blue") +
labs(title = "Life expectancy in white men in California, 1969-2013",
    y = "Life expectancy",
    x = "Year",
    caption = "Data from Riddell et al. (2018)")
```

# Line plot: life expectancy over time



# Relationships between two quantitative variables

# Explanatory and response variables





Bi-directional:

- "X predicts Y", or "Y predicts X"
- "X is associated with Y", or "Y is associated with X"

Unidirectional:

- "X causes Y"

# Which variable is x and which is y?

In prediction we generally use X to denote the variable we are using to predict the variable of interest (Y)

In causation we generally use X to denote the explanatory (independent) and Y to denote the response (dependent)

Graphically the X variable is on the X (horizontal) axis and the Y variable is the Y (vertical) axis

# Practice: choosing x and y

1. Each hospital's rate of hospital-acquired infections, and whether the hospital has implemented a hand-washing intervention as part of a cluster randomized trial.
2. The weight in kilograms and height in centimeters of a person
3. Inches of rain in the growing season and the yield of corn in bushels per day
4. A person's leg length and arm length, in centimeters

# How to investigate causation?

- Randomized controlled trials (RCTs) to randomize individuals to different levels
- Observational study that is *designed* to investigate causation and reduce the risk of bias

# Looking at relationships visually: Scatterplots

# Scatterplots

- Scatterplots are a good way to visualize a relationship between two variables
- When we look at a scatterplot we want to evaluate:
    - The overall Pattern of the dots
    - Any notable exceptions to the pattern
    - Direction (positive or negative)
    - Form (straight line or curved)
    - Strength (how closely the points follow a line)
    - Are there any obvious outliers

# Scatterplot syntax in R



```
name of plot <- ggplot(data = dataset, aes(x = xvariable, y = yvariable)) +
geom_point(na.rm=TRUE) + theme_minimal(base_size = 15)+
labs(x = "xlabel", y = "ylabel", title = "Title")
```

# NHANES data

Read in NHANES dataset

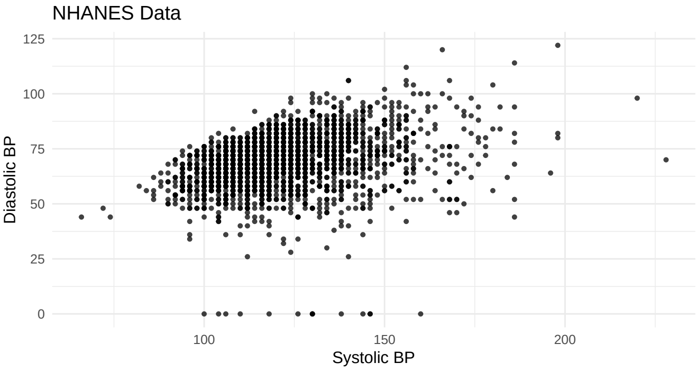
```
nhanes_dataNA <- read_csv("nhanes.csv")
nhanes_data<-nhanes_dataNA[rowSums(is.na(nhanes_dataNA[ , 15:18])) == 0, ]
nhanes_data %>%
  select(gender, bpxsy1, bpxdi1) %>%
  head()
```

```
## # A tibble: 6 x 3
##   gender bpxsy1 bpxdi1
##   <chr>   <dbl>  <dbl>
## 1 Male      140     90
## 2 Female    136     86
## 3 Female    118     80
## 4 Female    106     60
## 5 Female    122     56
## 6 Male      118     74
```

# Code: systolic and diastolic blood pressure

```r
nhanes_scatter <- ggplot(
  data = nhanes_data,
  aes(x = bpxsy1, y = bpxdi1)
) +
  geom_point(na.rm = TRUE, size = 0.9, alpha = 0.75) +
  theme_minimal(base_size = 10) +
  labs(x = "Systolic BP",
       y = "Diastolic BP",
       title = "NHANES Data")
```

# Scatterplot: systolic and diastolic blood pressure



# Discussion: what do we notice?

What do we notice from the plot?

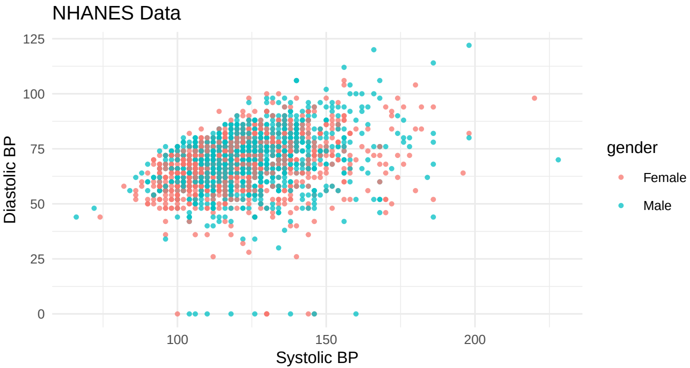
- Is there a visible association?
- Any notable exceptions to the pattern
- Direction (positive or negative)
- Form (straight line or curved)
- Strength (how closely the points follow a line)
- Are there any obvious outliers

# Adding a third variable with color

We can add a third variable to our graph by coloring the dots

```
nhanes_scatter <- ggplot(
  data = nhanes_data,
  aes(x = bpxsy1, y = bpxdi1)
) +
  geom_point(
    aes(col = gender),
    na.rm = TRUE,
    size = 0.9,
    alpha = 0.75
  ) +
  theme_minimal(base_size = 10) +
  labs(x = "Systolic BP",
       y = "Diastolic BP",
       title = "NHANES Data")
```

# Colored scatterplot: blood pressure by gender



# Manatee data



Manatee data set from your textbook:

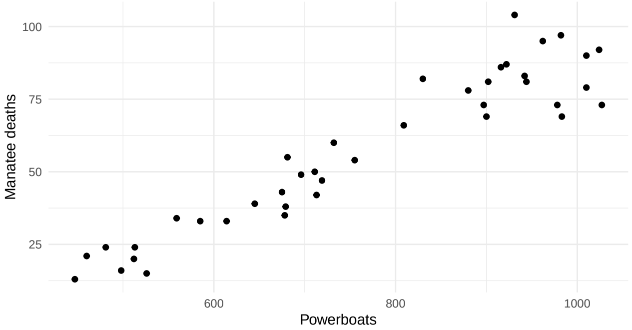
```
mana_data <- read_csv("Ch03_Manatee-deaths.csv")
head(mana_data)
```

```
## # A tibble: 6 x 3
##     year powerboats deaths
##    <dbl>      <dbl>  <dbl>
## 1  1977        447     13
## 2  1987        645     39
## 3  1997        755     54
## 4  2007       1027     73
## 5  1978        460     21
## 6  1988        675     43
```

# Code: powerboats and manatees

```
mana_scatter <- ggplot(
  data = mana_data,
  aes(x = powerboats, y = deaths)
) +
  geom_point(size = 1.5) + theme_minimal(base_size = 10) +
  labs(x = "Powerboats", y = "Manatee deaths")
```

# Scatterplot: powerboats and manatee deaths



# Discussion: manatee plot

What do we notice from the plot?

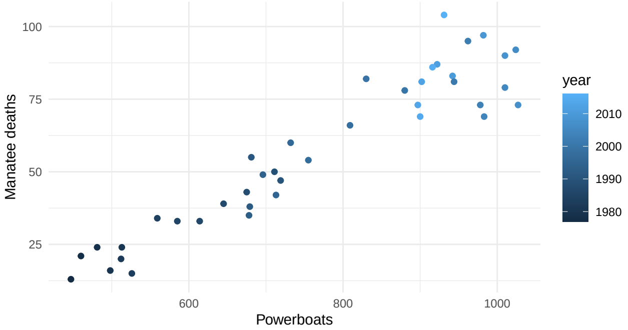
- Is there a visible association?
- Any notable exceptions to the pattern
- Direction (positive or negative)
- Form (straight line or curved)
- Strength (how closely the points follow a line)
- Are there any obvious outliers

# Adding year as a continuous third variable

What if we layer in a continuous third variable?

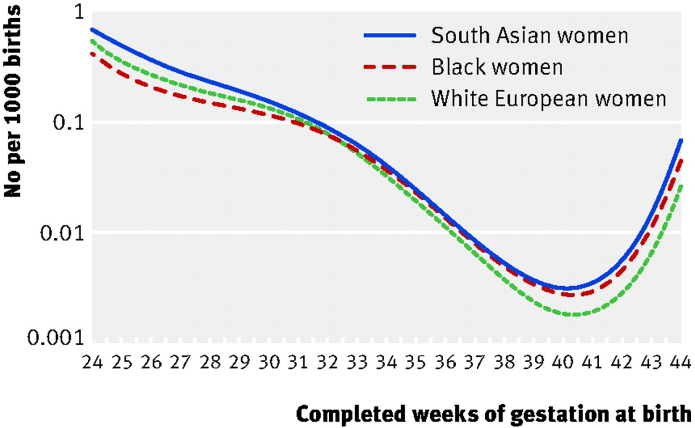
```r
mana_scatter <- ggplot(
  data = mana_data,
  aes(x = powerboats, y = deaths)
) +
  geom_point(aes(col = year), size = 1.5) +
  theme_minimal(base_size = 10) +
  theme(legend.position = "right") +
  labs(x = "Powerboats", y = "Manatee deaths")
```

# Manatee plot colored by year



# A non-linear example: gestational age and perinatal mortality

## Gestational age and perinatal mortality




Source: Balchin et al. BMJ. 2007.

# Exploratory analysis using scatterplots

# Lean body mass and metabolic rate: Problem and plan

Problem: Is lean body mass (person's weight after removing the fat) associated with metabolic rate (kilocalories burned in 24 hours)?

Plan: A diet study was conducted on 12 women and 7 men that measured lean body weight and metabolic rate for each individual.

# Lean body mass and metabolic rate: Data



Data: In the textbook

| Subject | Sex | Mass (kg) | Rate (Cal) | Subject | Sex | Mass (kg) | Rate (Cal) |
|---|---|---|---|---|---|---|---|
| 1 | M | 62.0 | 1792 | 11 | F | 40.3 | 1189 |
| 2 | M | 62.9 | 1666 | 12 | F | 33.1 | 913 |
| 3 | F | 36.1 | 995 | 13 | M | 51.9 | 1460 |
| 4 | F | 54.6 | 1425 | 14 | F | 42.4 | 1124 |
| 5 | F | 48.5 | 1396 | 15 | F | 34.5 | 1052 |
| 6 | F | 42.0 | 1418 | 16 | F | 51.1 | 1347 |
| 7 | M | 47.4 | 1362 | 17 | F | 41.2 | 1204 |
| 8 | F | 50.6 | 1502 | 18 | M | 51.9 | 1867 |
| 9 | F | 42.0 | 1256 | 19 | M | 46.9 | 1439 |
| 10 | M | 48.7 | 1614 | | | | |

What would the corresponding data frame look like? How many variables would it have? How many rows?

# Creating the weight data frame

Full dataset: 19 rows and 4 variables. Here are the first few rows:

```
head(weight_data, 4)
```

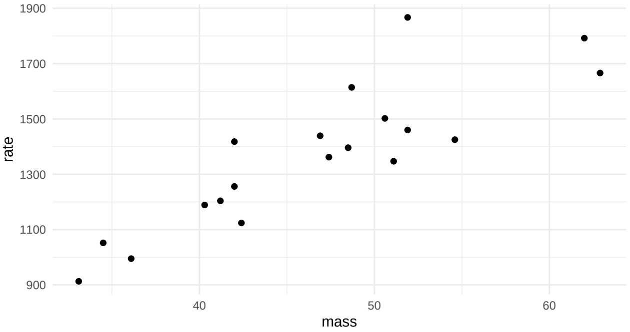
```
## # A tibble: 4 x 4
##   subject gender  mass  rate
##     <dbl> <chr>  <dbl> <dbl>
## 1       1 M       62    1792
## 2       2 M       62.9  1666
## 3       3 F       36.1   995
## 4       4 F       54.6  1425
```

# Code: scatterplot for weight data

Exploratory data analysis using scatter plots

```
weight_scatter <- ggplot(
  weight_data,
  aes(x = mass, y = rate)
) +
  geom_point(size = 1.5) +
  theme_minimal(base_size = 10)
```

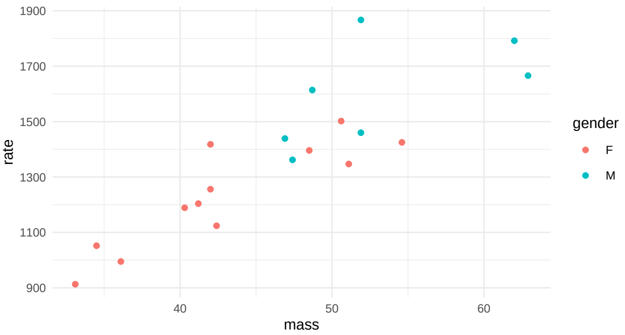
# Scatterplot: lean body mass and metabolic rate



# Code: color points by gender

```
weight_scatter <- ggplot(
  weight_data,
  aes(x = mass, y = rate)
) +
  geom_point(aes(col = gender), size = 1.5) +
  theme_minimal(base_size = 10)
```

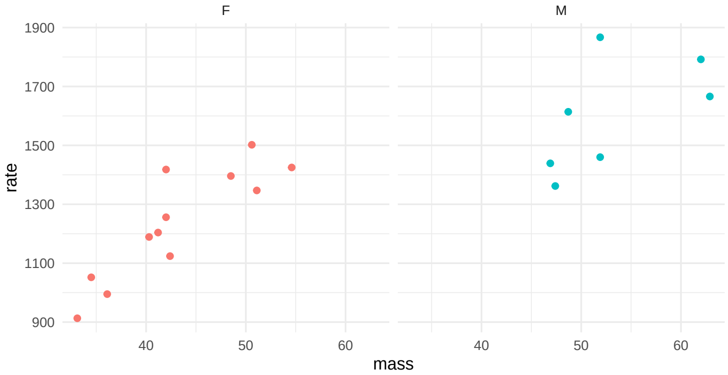
# Scatterplot colored by gender



# Code: separate plots for men and women

```
weight_scatter <- ggplot(
  weight_data,
  aes(x = mass, y = rate)
) +
  geom_point(aes(col = gender), size = 1.5) +
  theme_minimal(base_size = 10)+
  theme(legend.position = "none") +
  facet_wrap(~ gender)
```

# Faceted plots by gender



# Assessing a relationship between two variables with a number: Pearson's correlation

# Pearson's correlation: motivation

Using just our eyes, we can often say something about whether an association between two variables is weak or strong.

But we can also use a numeric value to describe the direction and strength of an association

# Pearson's correlation formula

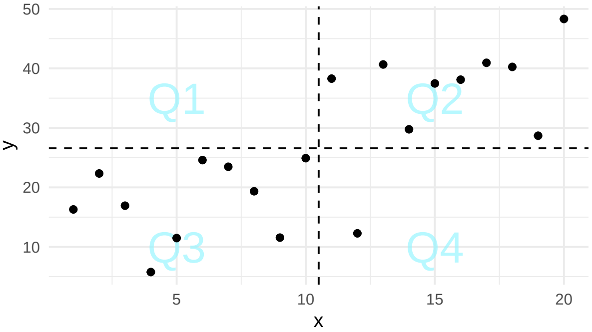
- For linear associations, we can use Pearson's correlation coefficient (denoted by $r$) to quantify the strength of a linear relationship between two variables.
- The correlation between $x$ and $y$ is:

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

Notice that because we are dividing by the standard deviation the values become unitless

# Intuition about correlation: center the variables

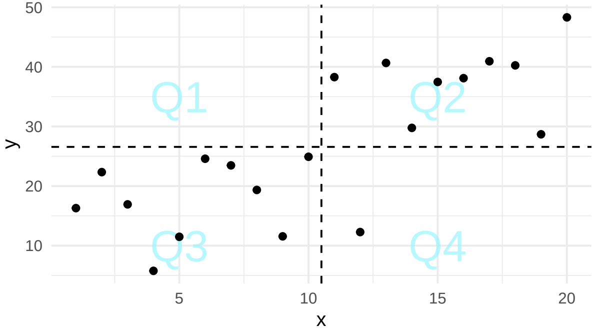
- First compare each value to its own mean: $x_i - \bar{x}$ and $y_i - \bar{y}$.
- The dashed lines split the plot into four regions.
- Each point is above or below the average value of each variable.



# Intuition about correlation: signs

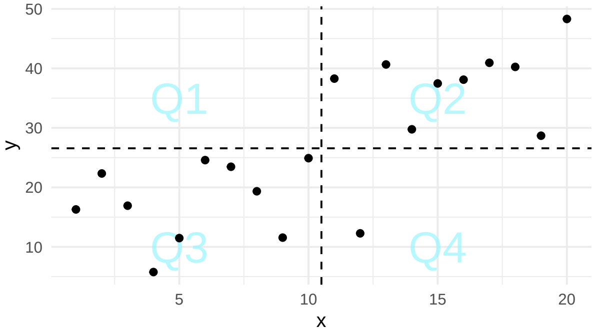
$$r = \frac{1}{n-1} \sum_{i=1}^{n} z_{x,i} z_{y,i}$$

- Same sign deviations give a positive contribution.
- Opposite sign deviations give a negative contribution.



# Intuition about correlation: direction

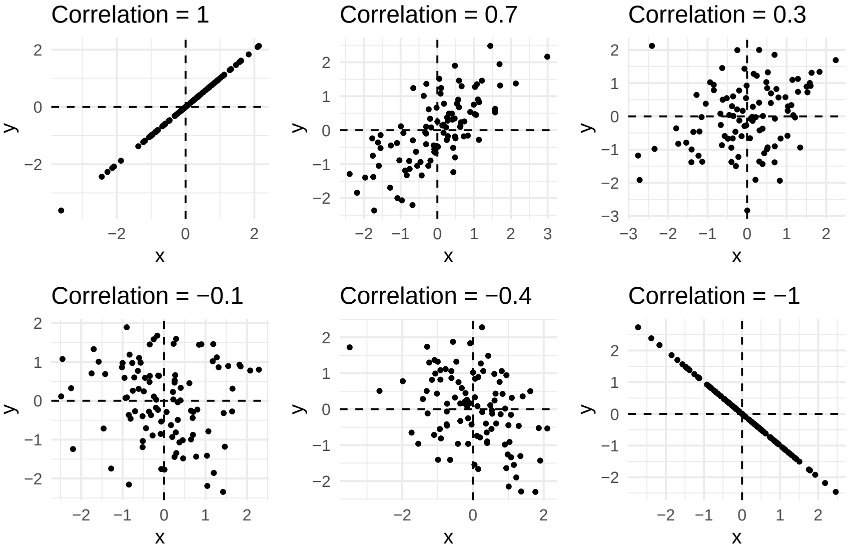
- More points in Q2 and Q3 $\Rightarrow$ positive correlation.
- More points in Q1 and Q4 $\Rightarrow$ negative correlation.
- A balance across regions $\Rightarrow$ correlation near 0.



# Properties of the correlation coefficient

- Always a number between -1 and 1.
  - -1: A perfect, negative linear association
  - 1: A perfect, positive linear association
  - 0: No linear association
- Is used to measure the association between two *quantitative* variables.
- Only useful for *linear* associations!

# Correlation and direction



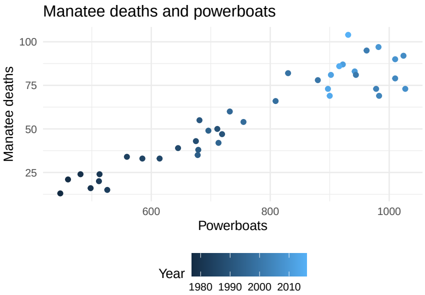
# Syntax: Pearson's correlation using `cor()`

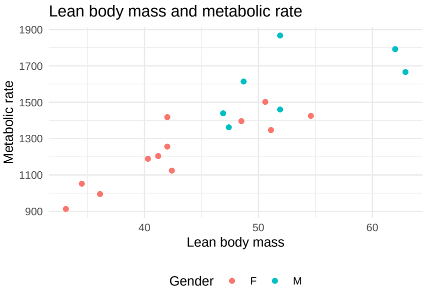
correlation coefficient <- dataset %>%

summarize(*newvar* = cor(xvar, yvar))

# Remember the manatee and weight plots

Before calculating $r$, predict the sign and rough strength from the plots.



# Correlation for manatee data

Now, calculate the correlations between X and Y for manatees:

```
mana_cor <- mana_data %>%
  summarize(corr_mana = cor(powerboats, deaths))
mana_cor
```

```
## # A tibble: 1 x 1
##   corr_mana
##       <dbl>
## 1     0.945
```

# Correlation for weight data

And for the weight data:

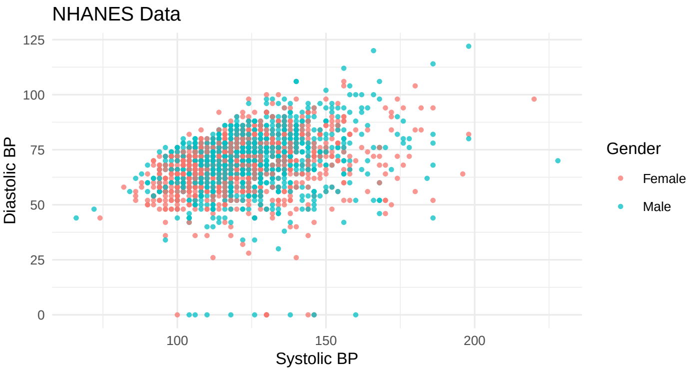
```
weight_cor <- weight_data %>%
  summarize(corr_weight = cor(mass, rate))
weight_cor
```

```
## # A tibble: 1 x 1
##   corr_weight
##         <dbl>
## 1       0.865
```

# Blood pressure plot before correlation



What about our blood pressure data from NHANES?



# Correlation for blood pressure data

```
bp_cor <- nhanes_data %>%
  summarize(corrbp = cor(bpxsy1, bpxdi1))
bp_cor
```

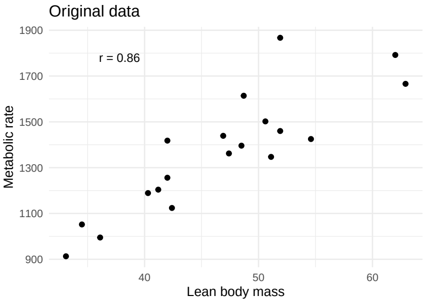
```
## # A tibble: 1 x 1
##   corrbp
##    <dbl>
## 1  0.322
```

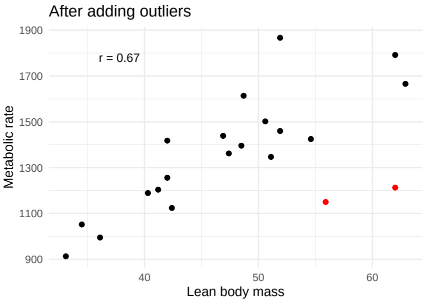
# Correlation is not resistant to outliers





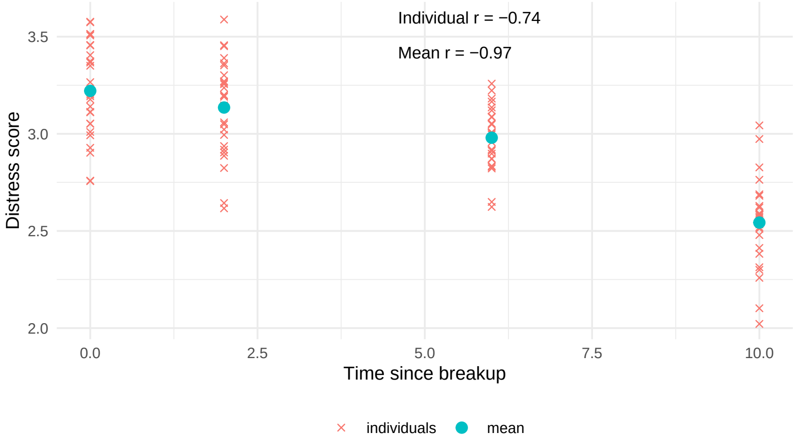
Adding two unusual points changes the correlation for the weight data.



# Correlations for averages versus individuals

- Correlations based on averages are often stronger than correlations based on individual data.



# Important concepts

- Determine which variable is explanatory and which is response, or when it doesn't matter
- Visually describe the relationship between two variables (form, direction, strength, and outliers)
- Numerically describe the relationship with the correlation coefficient $r$

# R Recap: What functions did we use?

- `geom_point()`
- `geom_line()`
- `aes(col = gender)` to color points by levels of `gender`
- `summarize()` to calculate correlation using `cor(var1, var2)`

# Reminder: association does not equal causation

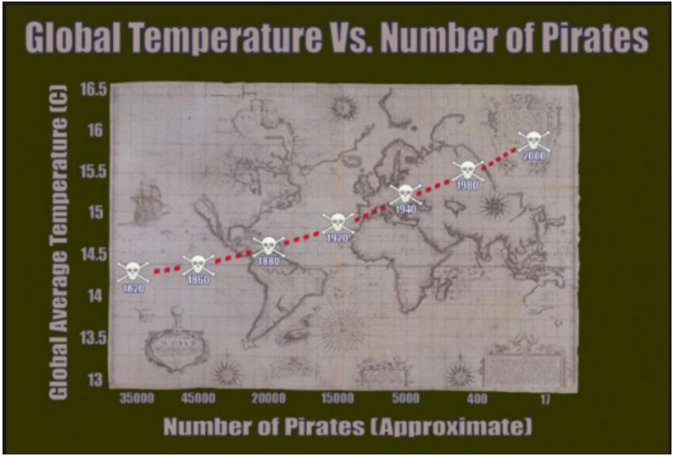


Remember that just because two variables are associated, does not mean there is a causal relationship

The correlation coefficient measures association *not* causation.

Even a very strong association doesn't mean that one variable causes the other.

# Reminder: association does not equal causation, pirate example




Example: a strong association can still be non-causal.